CS 201, Fall 2016 — Oct 28th — Exam 2 Name: _____

Question 1. [5 points] Consider the following static method:

```
public static int mystery(int[] arr) {
    int x = 0;
    for (int i = 0; i < arr.length; i++) {
        x += arr[i];
    }
    for (int i = 0; i < arr.length; i++) {
        x += arr[i];
    }
    return x;
}
```

Let N be the number of elements in the array `arr`. State a worst-case big-O upper bound on the running time of this method in terms of N. Explain your answer briefly.

Question 2. [5 points] Consider the following static method:

```
public static int numNegativeRows(int[][] matrix) {
    int count = 0;
    for (int i = 0; i < matrix.length; i++) {
        int rowSum = 0;
        for (int j = 0; j < matrix[i].length; j++) {
            rowSum += matrix[i][j];
        }
        if (rowSum < 0) { count++; }
    }
    return count;
}
```

Assume that the array `matrix` is square (has the same number of rows and columns), and that N is the number of rows/columns. State a worst-case big-O upper bound on the running time of this method in terms of N. Explain your answer briefly.

Question 3. [5 points] State a big-O upper bound on the worst-case running time of the following method, where the problem size N is the number of rows/cols in the array `mat`. You may assume the array has the same number of rows as columns. Briefly justify your answer, by indicating the approximate *average* number of iterations for each loop.

```
public static double diagSum(double[][] mat) {
  double sum = 0.0;
  int N = mat.length;
  for (int i = 1; i < N; i *= 2) {
    sum += mat[i][i];
  }
  return sum;
}
```

Question 4. [5 points] Consider the following static method:

```
public static boolean isUpperTriangular(double[][] matrix) {
    for (int i = 0; i < matrix.length; i++) {
        for (int j = 0; j < i; j++) {
            if (matrix[i][j] != 0.0) {
                return false;
            }
        }
    }
    return true;
}
```

Assume that the array `matrix` is square (has the same number of rows and columns), and that N is the number of rows/columns. State a worst-case big-O upper bound on the running time of this method in terms of N. Explain your answer briefly.

Question 5. [5 points] Consider the following static method:

```
public static boolean checkMatrix(double[][] mat) {
    int N = mat.length;
    for (int i = 1; i < N; i *= 2) {
        for (int j = 0; j < i; j++) {
            if (mat[i][j] < 0) {
                return false;
            }
        }
    }
    return true;
}
```

Assume that the array `matrix` is square (has the same number of rows and columns), and that N is the number of rows/columns. State a worst-case big-O upper bound on the running time of this method in terms of N. Explain your answer briefly.